Lesson 2: Valuation of Common Stocks

Module : Valuation of Fixed Income Securities and Stocks

Introduction

In this lesson we will cover the following topics:

- Introduction to common stocks and trading operations
- Fundamental principles of stock valuation
- Single-period and multi-period DCF approaches to stock valuation
- Dividend discount model and cost of equity
- Summary and concluding remarks

Trading of Securities on Exchanges

Trading of Securities on Exchanges

Let us discuss the basics of trading operations and the functioning of exchanges

- A large company such as Amazon has millions of shares being traded, and market capitalization that runs in trillions
- In such firms, investors comprise a number of small investors and a small group of large institutional investors such as pension funds and insurance companies
- If Amazon wants to raise more capital, it can do so by borrowing or selling new shares to investors
- This issuance of new shares to raise capital occurs in primary markets
- The trading of these already issued shares takes place in secondary markets

Trading of Securities on Exchanges

At these exchanges, investors buy and sell shares that are listed on the exchange

- This kind of secondary trading transactions only transfer the ownership from one owner to the other; no new shares are created
- The modern day electronic trading happens through two broad category of orders, often referred to as market orders and limit orders
- Market order is an order to trade immediately at the best available prices
- Limit orders state a price limit for execution ("worst available terms of trade"), if that price limit is met then only the order is executed; till then the limit order is stored in the exchange's limit order book

Trading of Securities on Exchanges

This buying and selling activity reflects the approval or disapproval of shareholders on the actions of firm managers, and therefore, has an indirect impact

- The present day trading requires these shares to be stored in the dematerialized (DEMAT) form
- Most of the modern day markets are auction markets where the auction is conducted by computers
- In addition to auction markets, some markets are called dealer-broker markets
- In dealer-broker markets, there is a designated market maker, who is willing to buy and sell large volumes of specific securities at the exchange
- This market maker ensures smooth, orderly, and continuous trading at the exchange
- Exchanges also provide summary of daily trading data on there website for wider information dissemination and price discovery

Market Value of Common Stocks

Market Value of Common Stocks

Every quarter, listed companies release their assets and liabilities in the form of balance-sheet and profit & loss statements

- These items are provided at book values, following international accounting norms
- These accounting numbers are based on opinions of auditors
- While the book values are very definite and tangible figures, they reflect historical value or cost of acquisition reducing depreciation
- This may not be an accurate representation of what these assets are worth today
- The market prices of firm stocks are often more than their book values, indicating that the firm has been taking positive NPV projects

Market Value of Common Stocks

Market prices of stocks can be more or less than book values, depending upon whether the firm is taking positive or negative NPV projects

- Financial analysts often argue that markets are efficient, and market prices are the best indicator of firm value
- However, very often, the projects, business, or companies to be valued are not publically traded

	Market-to-Book-Value Ratio		Price–Earnings Ratio	
	Company	Competitors*	Company	Competitors*
Johnson & Johnson	3.4	3.0	11.3	10.9
PepsiCo	6.4	3.0	15.6	12.9
Campbell Soup	9.0	4.6	8.8	11.3
Wal-Mart	3.0	2.1	14.6	13.4
Exxon Mobil	2.9	1.2	7.6	5.3
Dow Chemical	0.5	3.0	12.5	10.6
Dell Computer	4.5	3.7	7.9	11.1
Amazon	11.2	2.7	46.9	22.2
McDonald's	4.4	3.1	14.1	13.6
American Electric Power	1.1	1.1	8.1	11.0
GE	1.0	1.7	4.6	8.8

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions. Chapter 4

Market Value of Common Stocks

Investors value such companies and projects by looking at publically traded firms with similar risk profile to identify how much investors are willing to pay for each dollar of the earnings or assets of these firms

- Market to book ratio: i.e., market value of equity divided by the book value of equity.
- PE ratio: Share price divided by Earnings reported in the Profit and Loss statement.

	Market-to-Book-Value Ratio		Price–Earnings Ratio	
	Company	Competitors*	Company	Competitors*
Johnson & Johnson	3.4	3.0	11.3	10.9
PepsiCo	6.4	3.0	15.6	12.9
Campbell Soup	9.0	4.6	8.8	11.3
Wal-Mart	3.0	2.1	14.6	13.4
Exxon Mobil	2.9	1.2	7.6	5.3
Dow Chemical	0.5	3.0	12.5	10.6
Dell Computer	4.5	3.7	7.9	11.1
Amazon	11.2	2.7	46.9	22.2
McDonald's	4.4	3.1	14.1	13.6
American Electric Power	1.1	1.1	8.1	11.0
GE	1.0	1.7	4.6	8.8

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions. Chapter 4

Market Value of Common Stocks

Consider the hypothetical values of Market to Book ratio and PE ratio provided here

- The values are provided for firms and the average values for competitors in their industry
- For unlisted company valuation, you would rely on industry averages

	Market-to-Book-Value Ratio		Price–Earnings Ratio	
	Company	Competitors*	Company	Competitors*
Johnson & Johnson	3.4	3.0	11.3	10.9
PepsiCo	6.4	3.0	15.6	12.9
Campbell Soup	9.0	4.6	8.8	11.3
Wal-Mart	3.0	2.1	14.6	13.4
Exxon Mobil	2.9	1.2	7.6	5.3
Dow Chemical	0.5	3.0	12.5	10.6
Dell Computer	4.5	3.7	7.9	11.1
Amazon	11.2	2.7	46.9	22.2
McDonald's	4.4	3.1	14.1	13.6
American Electric Power	1.1	1.1	8.1	11.0
GE	1.0	1.7	4.6	8.8

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions. Chapter 4

Market Value of Common Stocks

It is no guarantee that such relative valuation will match with the actual figures available for traded peers

- However, such estimates provide comparable benchmarks
- These can also be employed to compare the discounted cash flows with the current market view about the industry sector

	Market-to-Book-Value Ratio		Price–Earnings Ratio	
	Company	Competitors*	Company	Competitors*
Johnson & Johnson	3.4	3.0	11.3	10.9
PepsiCo	6.4	3.0	15.6	12.9
Campbell Soup	9.0	4.6	8.8	11.3
Wal-Mart	3.0	2.1	14.6	13.4
Exxon Mobil	2.9	1.2	7.6	5.3
Dow Chemical	0.5	3.0	12.5	10.6
Dell Computer	4.5	3.7	7.9	11.1
Amazon	11.2	2.7	46.9	22.2
McDonald's	4.4	3.1	14.1	13.6
American Electric Power	1.1	1.1	8.1	11.0
GE	1.0	1.7	4.6	8.8

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions. Chapter 4

Fundamental Valuation of Stocks-I

Fundamental Valuation of Stocks-I

The comparable valuation method provides estimates of value that are more aligned to the current market expectations

- However, there is no surety that these expectations are accurate, and one may have to also fall back upon the fundamental valuation method
- Fundamental analysis of securities require knowledge of factors that affect security prices
- Fundamental analysis with discounted cash flow method can be applied to bonds (using coupon and principal cash flows) and stocks (using dividends) both
- These cash flows are essentially the expected future income from these securities: PV of Stock = PV of expected future dividend income

Fundamental Valuation of Stocks-I

Let us understand this through an example

- Your pay-off from investment in a stock is in two forms: (a) You expect to receive dividends; (b) you also expect capital gains (or losses)

- You bought the stock at P_0, you plan to sell it next year at P_1 and also expect to receive a dividend (DIV_1)

- Assuming no taxes, Expected Returns= $r = \frac{DIV_1 + P_1 - P_0}{P_0}$

- The actual price may be different from this expected price; 'r' here is the market capitalization rate or cost of equity capital

Fundamental Valuation of Stocks-I

Consider a company ABC with the current price P_0= \$100; dividend $DIV_1 = \$5$, and an expected price of \$110 at the end of the year. Then the expected returns by shareholders would be computed as shown here

- $r = \frac{DIV_1 + P_1 - P_0}{P_0} = \frac{5+110-100}{100} = 15\%$

- Assume that 15% is the interest that you (or other investors) expect from this stock and other stocks with similar risk

- $P_0 = \frac{DIV_1 + P_1}{1+r} = \frac{5+110}{1.15} = \$100$

Fundamental Valuation of Stocks-I

In the previous example, we computed the $\$100$ price based on the expectations of the stock price one year from today and discount rate based on the risk of the stock

- The actual price at which this stock is trading may not be the same, and may be more or less
- However, if you truly believe in your analysis, then investors like you would buy the undervalued stock and sell the overvalued stock
- As more and more investors will trade like this, the price of the security will rise or fall depending upon the demand supply dynamics
- In this manner, the prices will align to their efficient values

Fundamental Valuation of Stocks-II

Fundamental Valuation of Stocks-II

Let us examine the price of stock next year P_1. Similar to P_0, we can also write this price P_1 in terms of dividend DIV_2, price P_2 and the discount rate r

- $P_1 = \frac{DIV_2 + P_2}{1+r}$

- Subsequently, we can also write the current price (P_0) in terms of dividends for next two years, DIV_1 and DIV_2, and second year price P_2

- $P_0 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{P_2}{(1+r)^2}$

Fundamental Valuation of Stocks-II

Let us further consider the previous example of the company ABC. The investors are expecting a dividend of $5.00 in year 1, $5.50 in year 2, and a price of $121 at the end of year 2

- $P_0 = ??$
- This expansion of current price (P_0) can be further extended to 3, 4 or more periods
- Consider a rather large horizon of H periods, and the expected price at the end of the horizon as P_H
- The current price can be easily expressed in terms of dividends received during this period and the price obtained at the end, i.e., P_H
- $\mathbf{P_0 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3} + \cdots + \frac{DIV_H + P_H}{(1+r)^H} = \sum_{t=1}^{H} \frac{DIV_t}{(1+r)^t} + \frac{P_H}{(1+r)^H}}$

Fundamental Valuation of Stocks-II

Let us further consider the previous example of the company ABC. The investors are expecting a dividend of $5.00 in year 1, $5.50 in year 2, and a price of $121 at the end of year 2

- $P_0 = \frac{5.00}{1.15} + \frac{5.50}{1.15^2} + \frac{121}{1.15^2} = 100$
- This expansion of current price (P_0) can be further extended to 3, 4 or more periods
- Consider a rather large horizon of H periods, and the expected price at the end of the horizon as P_H
- The current price can be easily expressed in terms of dividends received during this period and the price obtained at the end, i.e., P_H
- $\mathbf{P_0 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3} + \cdots + \frac{DIV_H + P_H}{(1+r)^H} = \sum_{t=1}^{H} \frac{DIV_t}{(1+r)^t} + \frac{P_H}{(1+r)^H}}$

Fundamental Valuation of Stocks-II

$$\mathbf{P_0 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3} + \cdots + \frac{DIV_H + P_H}{(1+r)^H} = \sum_{t=1}^{H} \frac{DIV_t}{(1+r)^t} + \frac{P_H}{(1+r)^H}}$$

- $\sum_{t=1}^{H} \frac{DIV_t}{(1+r)^t}$; this expression indicates the discounted sum (PVs) of dividends for years 1 to H
- The value of current share price (P_0) has two components
  1. The present value of dividends to be received over the investment horizon
  2. The present value of the sale price at the end of investment period

Fundamental Valuation of Stocks-II

For company ABC, let us consider a 100-period horizon with 10% growth in dividends and prices year-on-year

	Expected Future Values		Present Values		
1.HorizonPeriod (H)	2.Dividend (DIVt)	3.Price (Pt)	4.Cumulative Dividends	5.Future Price	6.Total
0	—	100	—	—	100
1	5.00	110	4.35	95.65	100
2	5.50	121	8.51	91.49	100
3	6.05	133.10	12.48	87.52	100
4	6.66	146.41	16.29	83.71	100
10	11.79	259.37	35.89	64.11	100
20	30.58	672.75	58.89	41.11	100
50	533.59	11,739.09	89.17	10.83	100
100	62,639.15	1,378,061.23	98.83	1.17	100

Fundamental Valuation of Stocks-II

Columns (2) and (3) represent the expected future values of dividends and prices. Column (4) represents the present value of the total expected dividends to be received if the stock is held till that year

	Expected Future Values		Present Values		
1.HorizonPeriod (H)	2.Dividend (DIVt)	3.Price (Pt)	4.Cumulative Dividends	5.Future Price	6.Total
0	—	100	—	—	100
1	5.00	110	4.35	95.65	100
2	5.50	121	8.51	91.49	100
3	6.05	133.10	12.48	87.52	100
4	6.66	146.41	16.29	83.71	100
10	11.79	259.37	35.89	64.11	100
20	30.58	672.75	58.89	41.11	100
50	533.59	11,739.09	89.17	10.83	100
100	62,639.15	1,378,061.23	98.83	1.17	100

Fundamental Valuation of Stocks-II

Column (5) represents the present value of the expected sale price for that year. The investor that holds the stock for 100-periods, will receive a total present value of $100. Out of this, $98.83 come from the dividends received over the investment period and $1.17 come from the sale of share

	Expected Future Values		Present Values		
1.HorizonPeriod (H)	2.Dividend (DIVt)	3.Price (Pt)	4.Cumulative Dividends	5.Future Price	6.Total
0	—	100	—	—	100
1	5.00	110	4.35	95.65	100
2	5.50	121	8.51	91.49	100
3	6.05	133.10	12.48	87.52	100
4	6.66	146.41	16.29	83.71	100
10	11.79	259.37	35.89	64.11	100
20	30.58	672.75	58.89	41.11	100
50	533.59	11,739.09	89.17	10.83	100
100	62,639.15	1,378,061.23	98.83	1.17	100

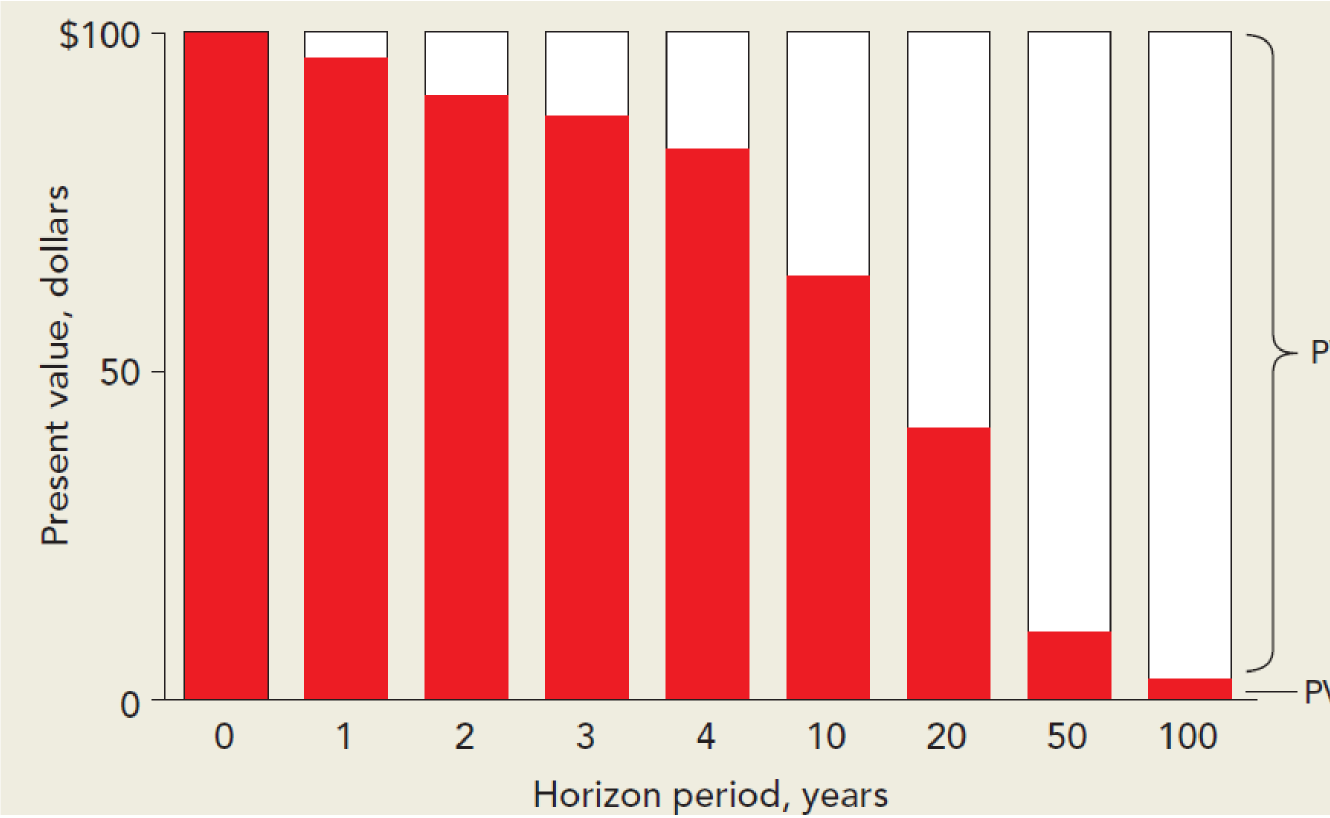
Fundamental Valuation of Stocks-II

For all periods, the total PV remains the same at $100. However, for investors with different horizons the composition of PV of dividend and sale price varies

	Expected Future Values		Present Values		
1.HorizonPeriod (H)	2.Dividend (DIVt)	3.Price (Pt)	4.Cumulative Dividends	5.Future Price	6.Total
0	—	100	—	—	100
1	5.00	110	4.35	95.65	100
2	5.50	121	8.51	91.49	100
3	6.05	133.10	12.48	87.52	100
4	6.66	146.41	16.29	83.71	100
10	11.79	259.37	35.89	64.11	100
20	30.58	672.75	58.89	41.11	100
50	533.59	11,739.09	89.17	10.83	100
100	62,639.15	1,378,061.23	98.83	1.17	100

Fundamental Valuation of Stocks-II

Let us examine these arguments with the help of the figure shown here. For investors with long horizons, the PV of dividends has the major share in the value of current price P_0

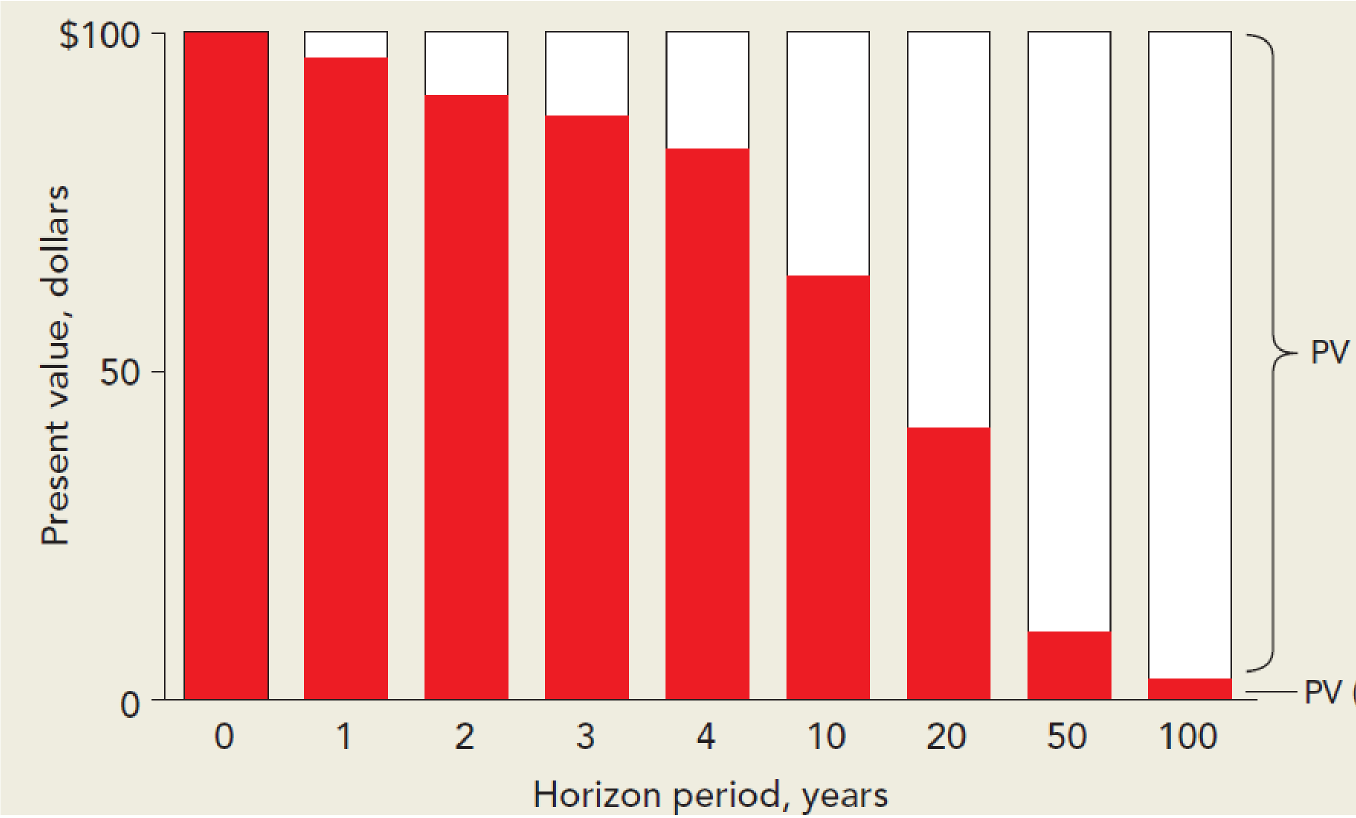


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

For the investor with small horizons, the major share of the current price comes from the sale price and a rather small share from the PV of dividends

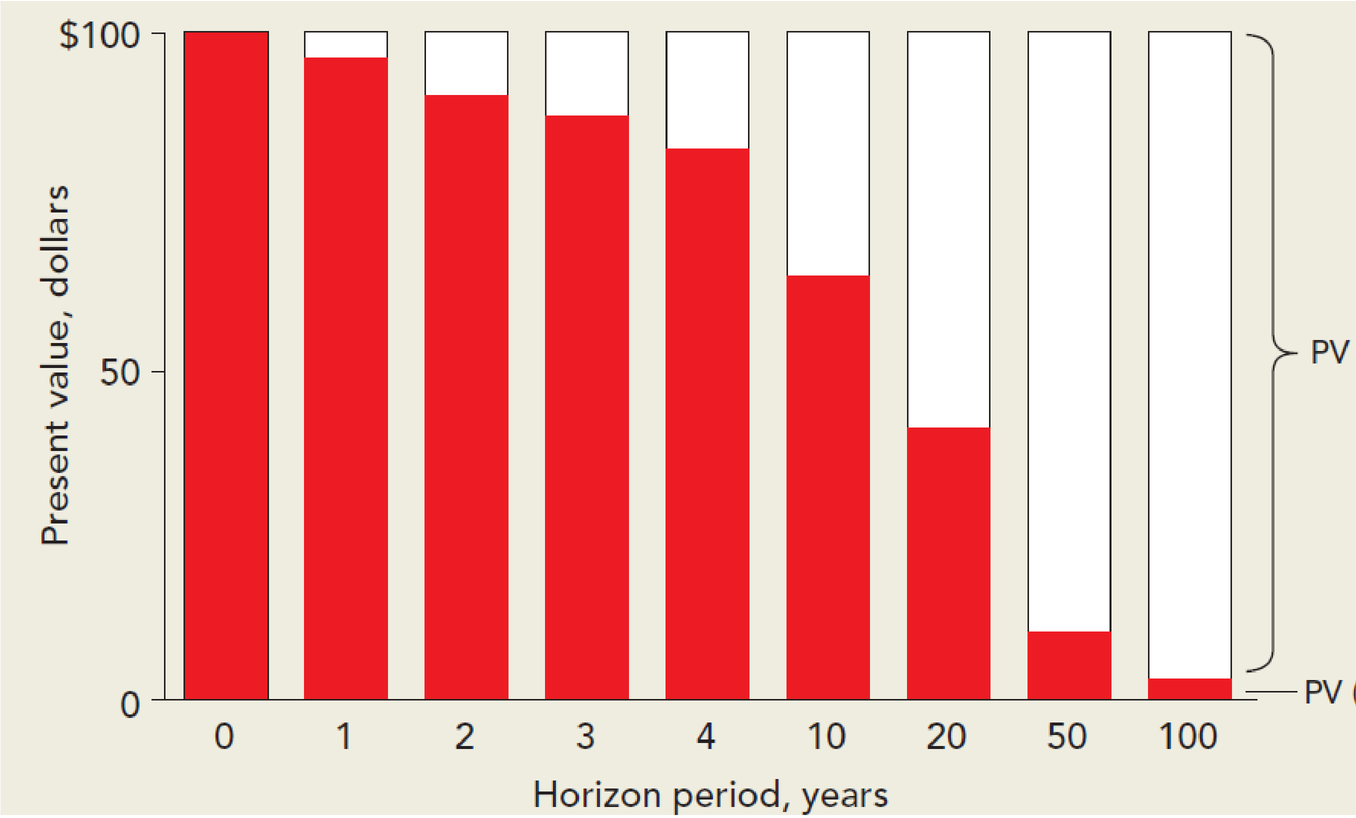


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

Essentially, even for short-horizon investors, the value of sale price is determined by expectations of dividends only (the selling price in any year is nothing but the PV of expected dividends in future)

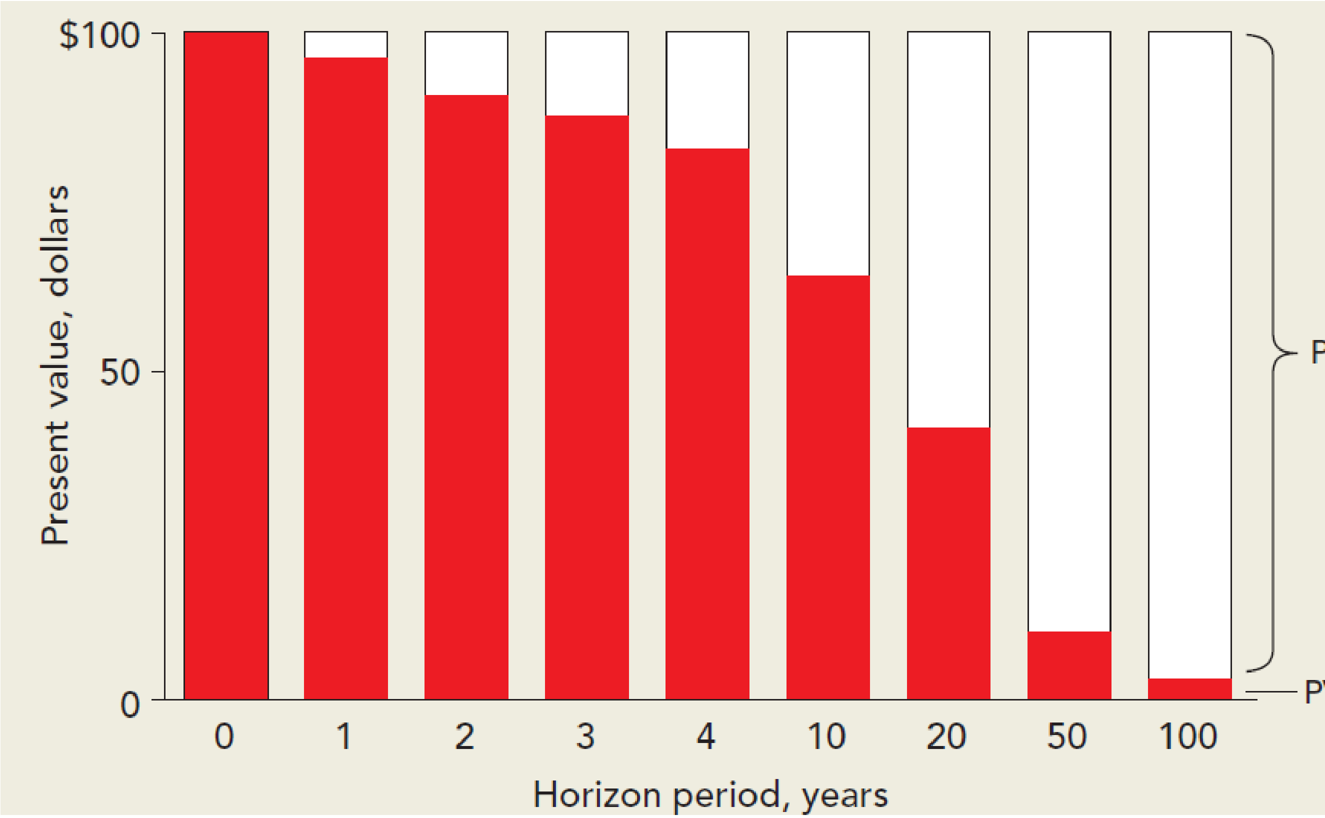


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

As the investment horizon increases and H becomes infinitely large, the present value of sale price approaches zero

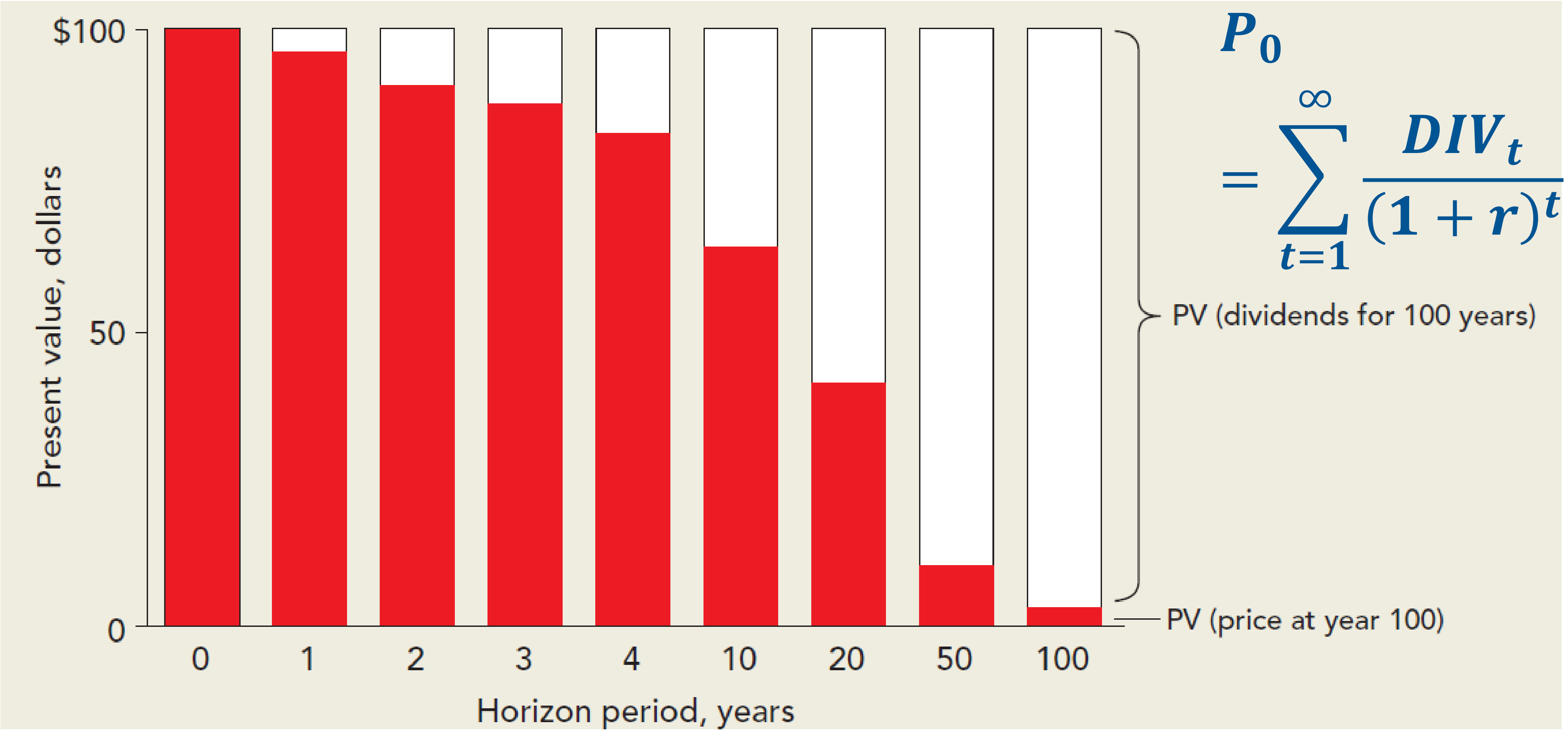


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

The assumption of infinitely large investment horizon in a stock is not unrealistic: separation of ownership and control

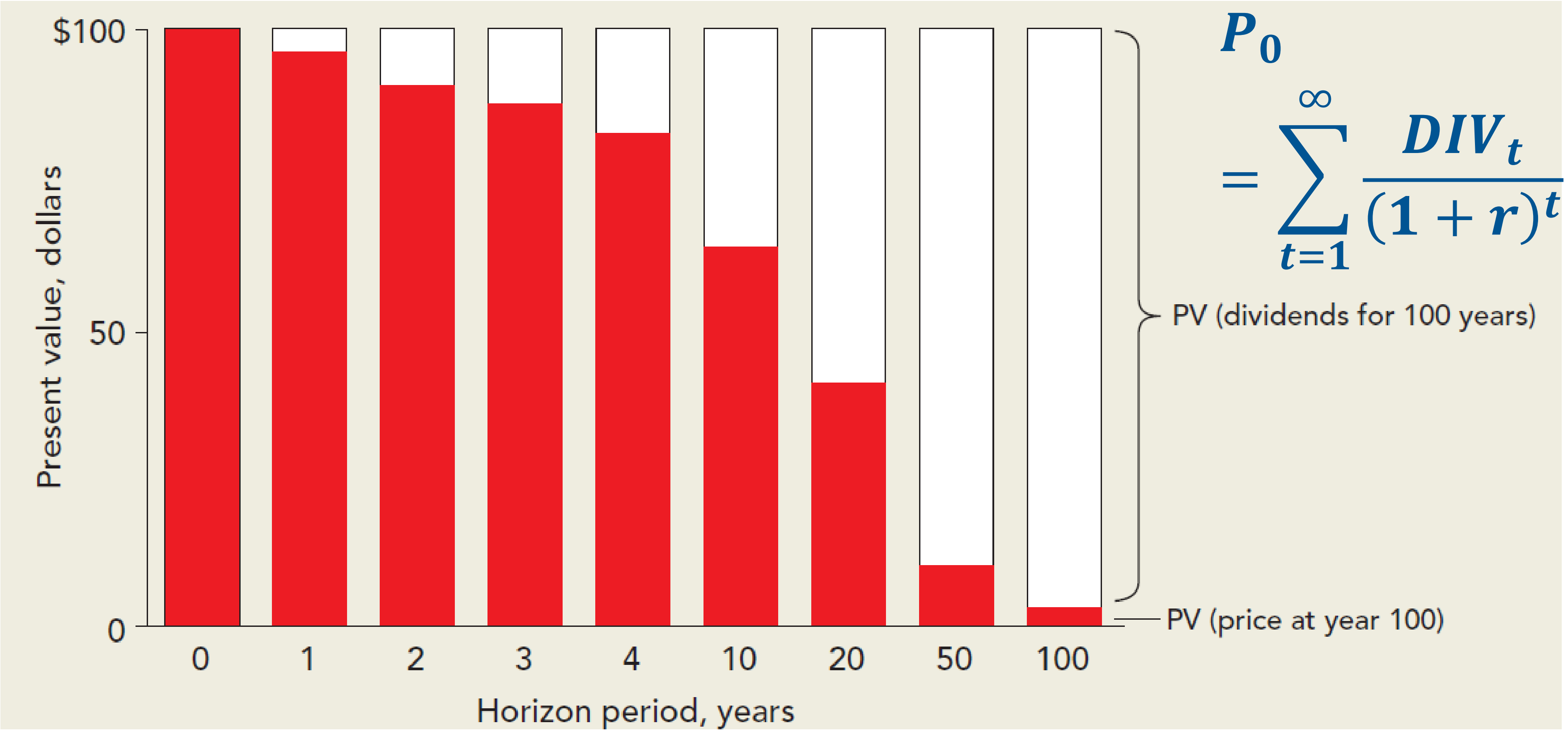


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

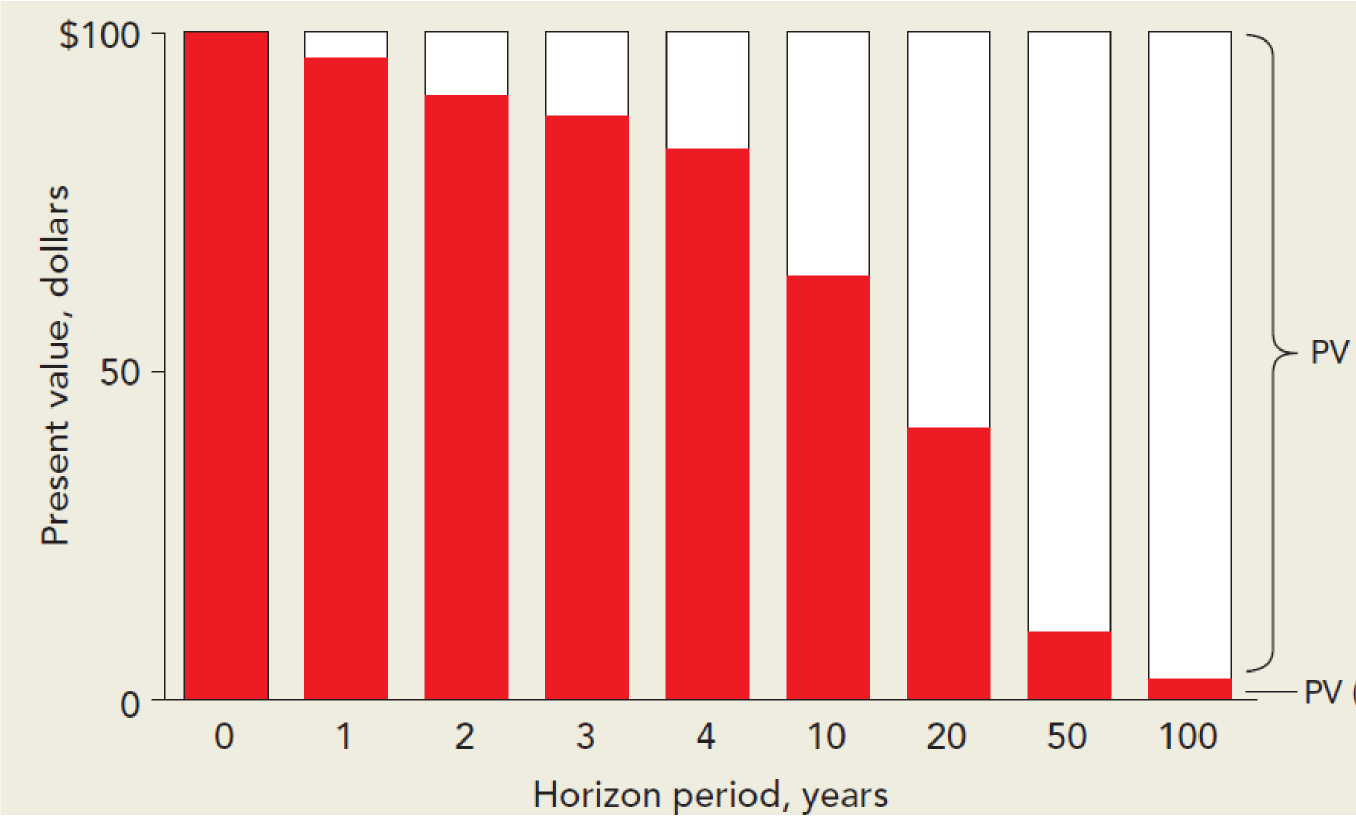
Therefore, our resulting expression for the current price of stock simplifies to: $P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$



Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

At times this formula of discounting dividends may not seem very intuitive as it does not factor the capital gains

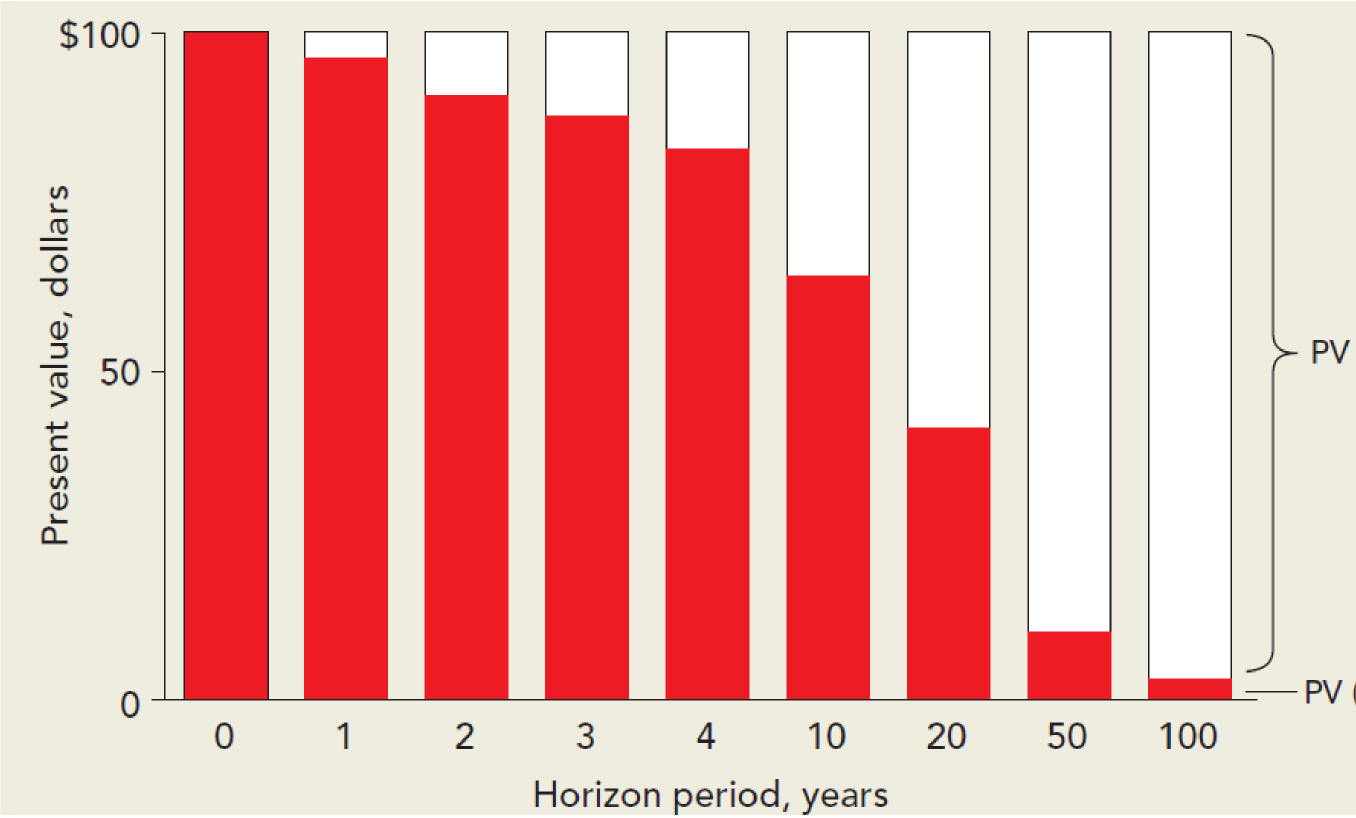


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

Future prices, that are used to compute capital gains, are essentially determined by the future expectations of dividends only (not earnings)

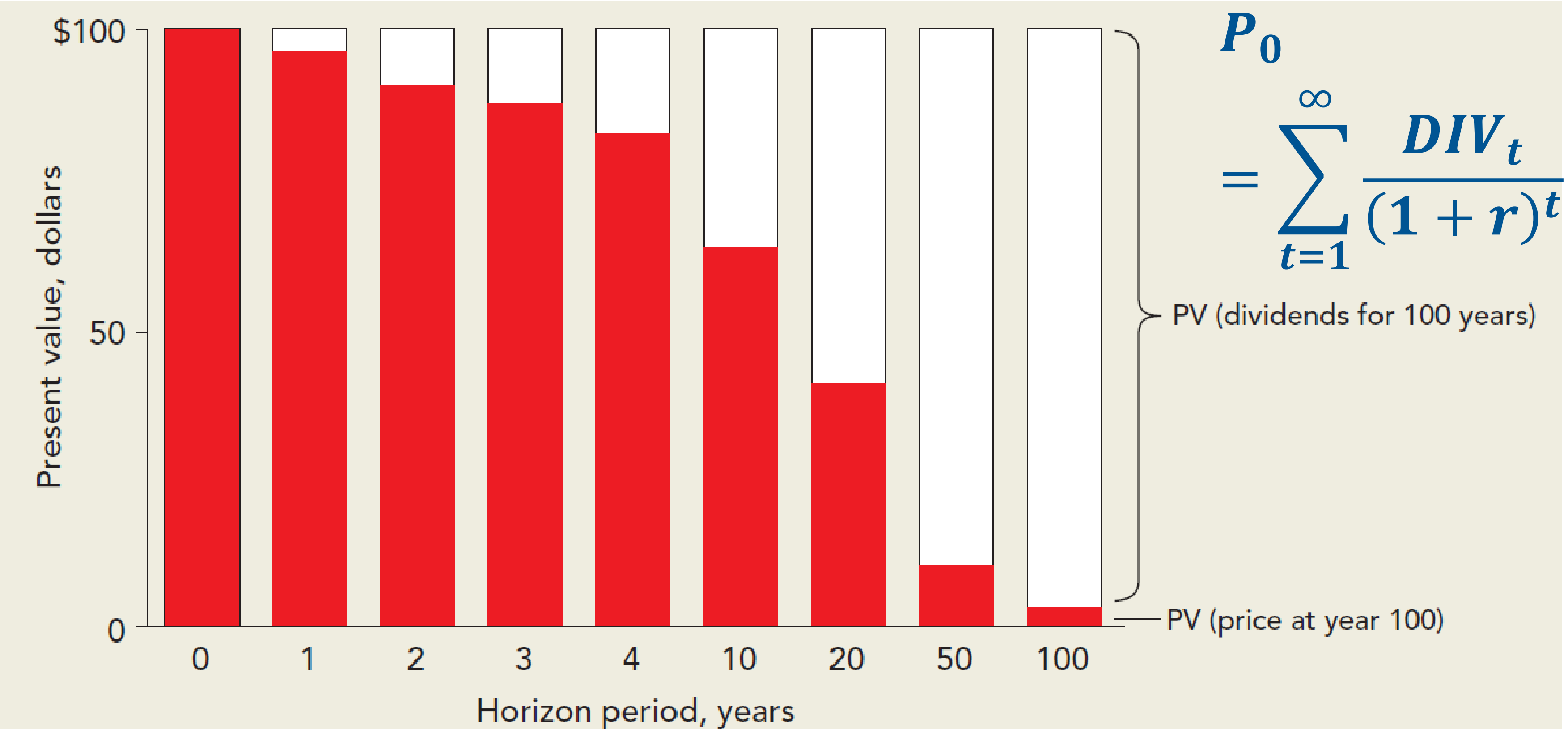


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

If one discounts the earnings, then it would be double-counting of cash inflows, as the investment into plant and machinery would also be included that has resulted in sustained (and at times increasing) dividends

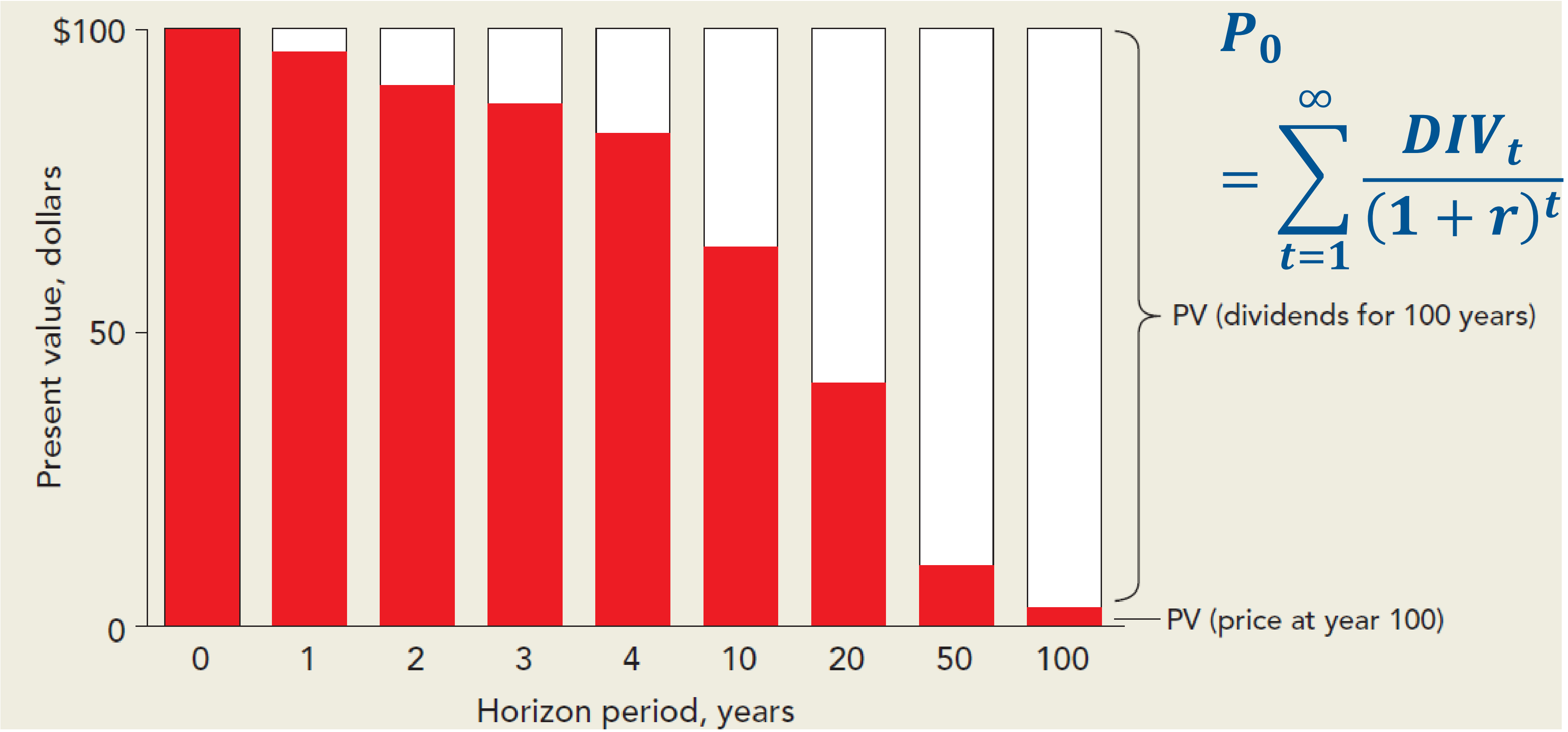


$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Fundamental Valuation of Stocks-II

However, valuing young growth companies that do not pay dividends or make losses is difficult using this model



$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

Brealey, Myers and Allen; Principles of Corporate Finance. 10th, 11th, or 12th editions, Chapter 4

Dividend Discount Model and Cost of Equity Capital

Dividend Discount Model and Cost of Equity Capital

Let us put our knowledge of DCF valuation through dividends to practice

- Assume a constant long-term growth rate of ‘g’ in dividends and an appropriate discount rate ‘r’
- Assuming a dividend of DIV_1 in the first year, this perpetuity can be valued using the formula shown

  here: $P_0 = \frac{DIV_1}{r-g}$

- Here, ‘g’ is the anticipated long-term growth rate, and is less than ‘r’

Dividend Discount Model and Cost of Equity Capital

This growing perpetuity formula explains the current price (P_0) in terms of expected dividend DIV_1, growth 'g' and expected return on the securities of the same risk 'r'

- If the current price (P_0) is observed, this formula can be used to estimate 'r' as shown here: $r = \frac{DIV_1}{P_0} + g$
- This formula shows that the discount rate 'r' (or expected returns) equals the dividend yield ($\frac{DIV_1}{P_0}$) and the expected rate of growth 'g'
- This 'r' is often referred to as the expected return or cost of equity

Dividend Discount Model and Cost of Equity Capital

Let us understand this through a simple example

- Firm XYZ has a share price of \$42.45 at the start of the period, expected dividends starting from the year end amount to \$1.68 per share, payout ratio is 60% (plowback = 40%)
- The dividend yield for this stock can be simply computed as Dividend yield= $\frac{DIV_1}{P_0} = \frac{1.68}{42.45} = 4.0\%$
- The second part, that is estimating long-term growth rate ‘g’ is rather difficult
- The growth rate is estimated with the payout ratio and return on equity

Dividend Discount Model and Cost of Equity Capital

Growth rate ‘g’= plowback ratio*ROE

- $\text{Plowback raio} = 1 - \text{payout ratio} = 1 - \frac{\text{DIV}}{\text{EPS}} = 1 - .60 = 0.40$
- If the ROE is 10% and plowback ratio is 40%, the growth rate ‘g’ can be calculated with this simple formula: g=40%*10%= 4%
- Then your overall estimate of cost of equity capital: r= $\frac{DIV_1}{P_0} + g = 4.0\% + 4.0\%$=8.0%
- Such estimates are often noisy and prone to errors of estimation
- Constant growth dividend discount formula employed earlier is extremely sensitive to changes in the values of ‘g’ and ‘r’

Dividend Discount Model and Cost of Equity Capital

To improve these estimates, it is advisable to use multiple companies and consider industry average estimates

- These averages help in reducing the noise component that is often associated with a single company estimate
- Before applying this formula, one may need to be sure that the current high rates of growth are sustainable in the long term or not
- The constant growth DCF formula assumes the growth ‘g’ to continue in perpetuity
- Organizations often witness periods of high-growth rates due to favorable operating environments, which may not be sustainable in long term

Dividend Discount Model and Cost of Equity Capital

Consider the example of a firm with equity of \$25, $DIV_1 = \$0.5$ and $P_0 = \$50.0$. The firm has an ROE of 25% and payout ratio of 20%, and we want to compute the cost of equity 'r' (i.e., expected returns)

- Let us first compute the cost of equity for this firm
- Dividend growth rate= (1-Payout ratio)*ROE= ?
- $r = Dividend\ yield + g = ?$

Dividend Discount Model and Cost of Equity Capital

Consider the example of a firm with $DIV_1 = \$0.5$ and $P_0 = \$50.0$. The firm has an ROE of 25% and payout ratio of 20%, and we want to compute the cost of equity 'r' (i.e., expected returns)

- Let us first compute the cost of equity for this firm
- Dividend growth rate= (1-Payout ratio)*ROE= (1-0.20)*0.25= 20%
- $r = Dividend\ yield + g = \frac{0.5}{50} + 20\% = 1\% + 20\% = 21\%$ (!! Doesn't seem to be a fair estimate)
- No firm can sustain a growth rate of 21% infinitely into future. In real life such growth rates drop gradually over the years and attain that lower long-term growth that is sustainable

Dividend Discount Model and Cost of Equity Capital

Let us see the timeline of cash flow profile as shown here.

- To simplify things here, assume that the firm ROE drops to 16% in the third year
- Also that the payout ratio increases to 50%
- So now we have new growth figure, i.e., g=?

Years	1	2	3	4
Equity	10.00	?	?	?
Return on equity, ROE	0.25	0.25	0.16	0.16
Earnings per share, EPS	?	?	?	?
Payout ratio	0.20	0.20	0.50	0.50
Next year growth= (1-Payout)*ROE	?	?	?	
Dividends per share, DIV	?	?	?	?

Dividend Discount Model and Cost of Equity Capital

Let us see the timeline of cash flow profile as shown here.

- To simplify things here, assume that the firm ROE drops to 16% in the third year
- Also that the payout ratio increases to 50%
- So now we have new growth figure, i.e., g= 0.50*16%= 8%

Years	1	2	3	4
Equity	10.00	10*1.20=12.00	12*1.20=14.40	14.40*1.08=15.55
Return on equity, ROE	0.25	0.25	0.16	0.16
Earnings per share, EPS	10*0.25=2.50	12*0.25=3.00	14.40*0.16=2.30	15.55*0.16=2.49
Payout ratio	0.20	0.20	0.50	0.50
Next year growth= (1-Payout)*ROE	(1-0.2)*0.25=0.20	(1-0.2)*0.25=0.20	(1-0.5)*0.16=0.08	(1-0.5)*0.16=0.08
Dividends per share, DIV	2.5*0.2=0.50	3*0.20=0.60	2.30*0.5=1.15	2.49*0.5=1.245

Dividend Discount Model and Cost of Equity Capital

In order to compute the current price (P_0), one needs to use the DCF formula in two stages

- First, the high-growth phase and second the steady state

- In the high-growth phase, we need to value the three dividend inflows in year 1, 2, and 3

- Present value of dividends obtained in years 1, 2, and 3: $\frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3}$

- Steady state growth phase with cash flows in perpetuity: $\frac{P_3}{(1+r)^3} = \frac{DIV_4}{(r-g)} * \frac{1}{(1+r)^3}$

Dividend Discount Model and Cost of Equity Capital

We also know that the current price is observed as \$50. The resulting expression is provided here

- $50 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3} + \frac{DIV_4}{(r-g)} * \frac{1}{(1+r)^3}$; or $50 = \frac{0.50}{1+r} + \frac{0.60}{(1+r)^2} + \frac{1.15}{(1+r)^3} + \frac{1.245}{(r-0.08)} * \frac{1}{(1+r)^3}$

- Solving for above equation, we get a value of r=??

Dividend Discount Model and Cost of Equity Capital

We also know that the current price is observed as \$50. The resulting expression is provided here

- $50 = \frac{DIV_1}{1+r} + \frac{DIV_2}{(1+r)^2} + \frac{DIV_3}{(1+r)^3} + \frac{DIV_4}{(r-g)} * \frac{1}{(1+r)^3}$; or $50 = \frac{0.50}{1+r} + \frac{0.60}{(1+r)^2} + \frac{1.15}{(1+r)^3} + \frac{1.245}{(r-0.08)} * \frac{1}{(1+r)^3}$
- Solving for above equation, we get a value of r=9.94%
- These PV computations employed a two-stage DCF valuation model
- In the first stage, the high-growth phase, the firm was highly profitable (ROE=25%), and it plowed back 80% of earnings

Dividend Discount Model and Cost of Equity Capital

Third year onwards, profitability declined and payout increased, i.e., dividends increased

- That is less money plowed back; the long-term growth settled at a steady rate of 8%
- This example can be suitably extended to three or more stages of different growth regimes
- For example, first phase at growth rate of 20%, second phase of 12%, and finally the steady state at growth rate of 8%
- In this case, the present value will be computed in three stages
- DCF valuation of stock may not be consistent with those observed in the financial markets
- This could be due to the fact that markets may not agree with your growth projections, dividend forecasts, or cost of capital estimates

Summary and Concluding Remarks

Summary and Concluding Remarks

The value of stock is equal to the discounted dividend payments expected to be received in perpetuity

- $PV = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$
- However, investors often do not plan to hold the stock for eternity and have finite investment horizons
- These investment horizons involve returns in the form of dividends and capital gains
- The value of the stocks with infinite stream of growing dividends: $P_0 = \frac{DIV_1}{r-g}$

Thanks!